

TELCO CHURN PREDICTION REPORT

Course: DSC3173 - Artificial Intelligence

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1. Introduction

This report presents a comparative analysis between two machine learning approaches for predicting customer churn in the Telco dataset: an **XGBoost Classifier** and a **Keras Multi-Layer Perceptron (MLP)** deep learning model.

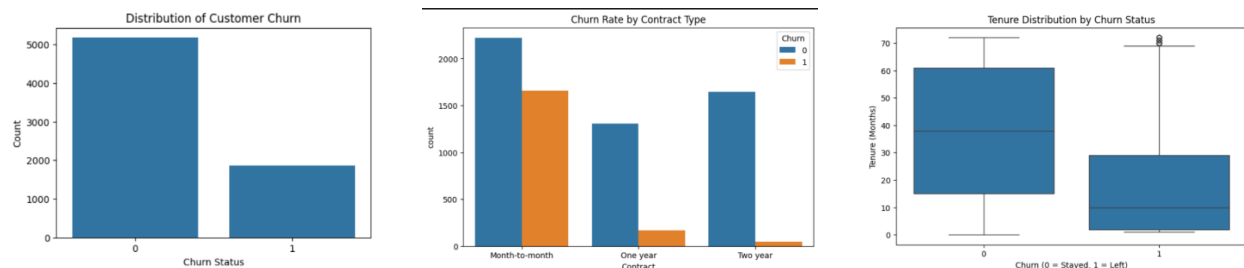
2. Overview of the Dataset

The dataset used in this study is the **Telco Customer Churn** dataset, which contains information about 7,043 customers of a telecommunications company in California. It includes detailed customer profiles across demographic, financial, and service usage parameters.

- **Total Records:** 7,043 customers
- **Total Features:** 20 predictor variables + 1 target variable (Churn)
- **Target Variable:** Churn (Binary: Yes / No) — indicating whether the customer left the company.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand feature distributions, identify data anomalies, and analyze relationships between predictor variables and customer churn.



Using plots to identify pattern in the dataset here I can identify some pattern between churn rate with contract type and Tenure

4. Data Preprocessing & Methodology

Before training the models, the dataset underwent several preprocessing steps:

- **Missing Value Imputation:** Handled missing or blank values in numerical columns like TotalCharges.
- **Categorical Encoding:** Converted categorical attributes into numerical formats using One-Hot Encoding.
- **Feature Scaling:** Applied standard scaling to normalize feature distributions for optimal neural network convergence.
- **Train/Test Split:** Partitioned the preprocessed dataset into training and testing sets.

5. Model Building & Hyperparameter Tuning

5.1 XGBoost Classifier

The XGBoost model was configured and optimized using hyperparameter tuning (e.g., controlling max_depth, learning_rate, and n_estimators) to establish a strong ensemble baseline.

5.2 Keras Multi-Layer Perceptron (MLP)

The Keras MLP model was built using the Sequential API with:

- **Architecture:** Input layer matching feature dimensions, 2 Dense hidden layers with ReLU activation, Batch Normalization, and Dropout layers to prevent overfitting.
- **Output Layer:** A single unit with Sigmoid activation for binary classification.
- **Hyperparameter Tuning:** Automated tuning using Keras Tuner (RandomSearch) to find optimal values for hidden units, dropout rates, and learning rates, targeting maximum validation AUC.

6. Performance Metrics Comparison

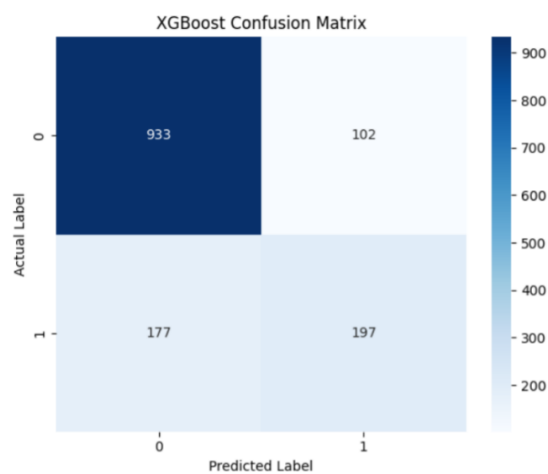
Both models were evaluated on the unseen test dataset across key performance indicators:

6.1 Performance Comparison Table

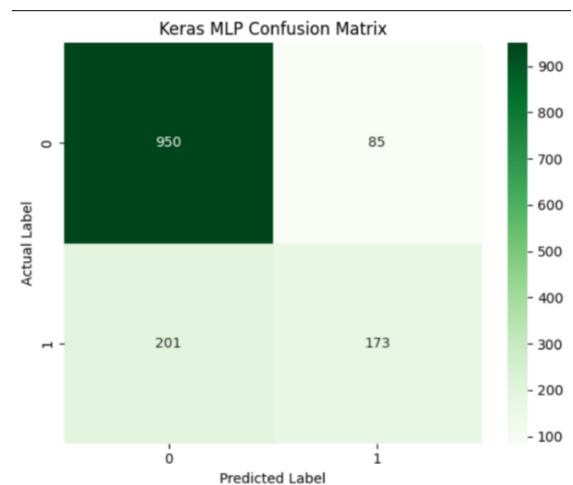
Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
XGBoost Classifier	<i>0.802</i>	<i>0.6589</i>	<i>0.5267</i>	<i>0.5854</i>	<i>0.8459</i>
Keras MLP	<i>0.797</i>	<i>0.6705</i>	<i>0.4626</i>	<i>0.5475</i>	<i>0.8411</i>

6.2 Visualizations

Confusion Matrices

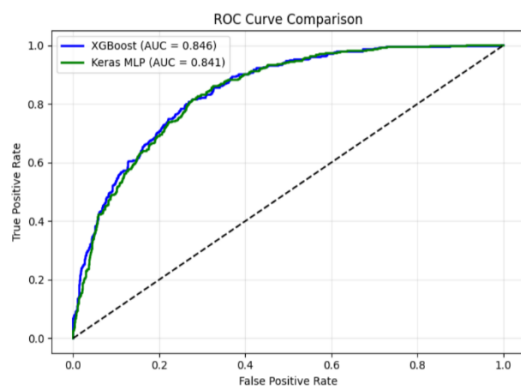


The XGBoost model correctly identified 197 churned customers (True Positives) and 933 non-churned customers (True Negatives).



The Keras MLP model correctly identified 173 churned customers (True Positives) and 950 non-churned customers (True Negatives).

ROC Curve Comparison

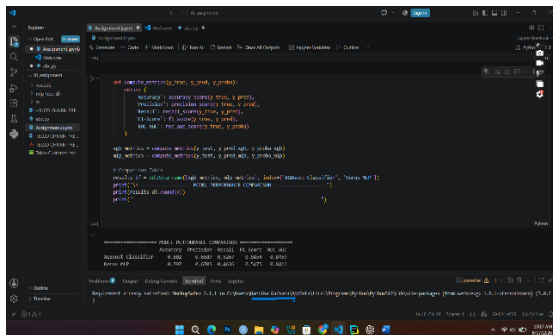


AUC Score Comparison: The XGBoost model achieved a slightly higher Area Under the Curve (AUC = 0.846) compared to the Keras MLP model (AUC = 0.841).

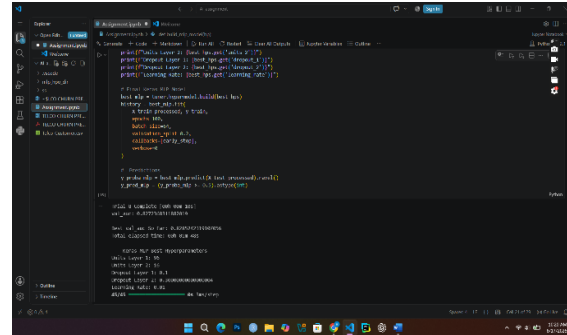
7. Discussion & Conclusion

- **Model Evaluation:** Both XGBoost and Keras MLP models successfully captured non-linear patterns within the Telco churn dataset. XGBoost easily captured complex patterns between combined features in tabular data and handled class imbalance efficiently, whereas the Keras MLP benefited from automated hyperparameter tuning and regularization layers to minimize overfitting.
- **Best Model Selection:** Based on the empirical evaluation metrics (specifically ROC-AUC of **0.8459**, F1-Score of **0.5854**, and Recall of **0.5267**), the **XGBoost Classifier** model demonstrated superior predictive capabilities compared to the Keras MLP.
- **Business & Practical Implications:** While Keras MLP achieved a slightly higher Precision (0.6705), XGBoost significantly outperformed it in Recall (0.5267 vs 0.4626). In customer churn prediction, higher Recall is critical as identifying actual churners (True Positives) allows the company to execute proactive retention strategies before customers leave. Therefore, **XGBoost Classifier is recommended as the optimal deployment model**.

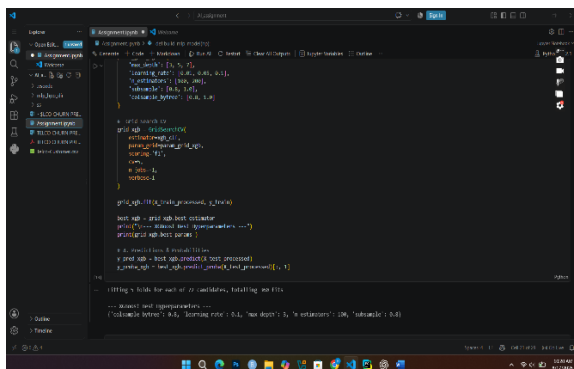
8. User verification screenshots



```
def model_performance(y_true, y_pred, y_prob):  
    accuracy = accuracy_score(y_true, y_pred)  
    precision = precision_score(y_true, y_pred)  
    recall = recall_score(y_true, y_pred)  
    f1_score = f1_score(y_true, y_pred)  
    auc_roc = roc_auc_score(y_true, y_prob)  
    return accuracy, precision, recall, f1_score, auc_roc  
  
# Hyperparameter Tuning  
best_auc = 0.0  
best_params = {}  
for param in param_grid:  
    model = XGBClassifier(**param)  
    model.fit(X_train, y_train)  
    y_prob = model.predict_proba(X_test)[0, :]  
    acc, prec, rec, f1, auc = model_performance(y_test, y_prob)  
    if auc > best_auc:  
        best_auc = auc  
        best_params = param  
  
# Final Model Evaluation  
model = XGBClassifier(**best_params)  
model.fit(X_train, y_train)  
y_prob = model.predict_proba(X_test)[0, :]  
acc, prec, rec, f1, auc = model_performance(y_test, y_prob)  
print(f"Final Model Performance: Accuracy: {acc}, Precision: {prec}, Recall: {rec}, F1-Score: {f1}, ROC-AUC: {auc}")
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